

SECTION AAnswer **all** questions.

1. Bromine water can be used to test for alkenes.

(a) (i) State the expected colour change for a positive test for alkenes. [1]

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(ii) Draw the structure of the product formed when propene reacts with bromine water. [1]

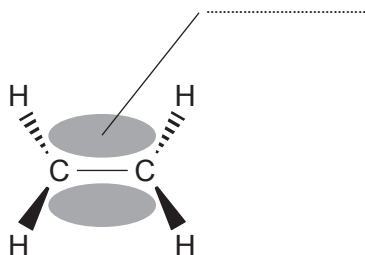
(b) Identify another reagent that can be used to test for the presence of alkenes. [1]

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2. Bonds in hydrocarbons are formed by the overlap of orbitals between each atom.

(a) Draw an *s*-orbital and a *p*-orbital in the space below. [1]

(b) Name the type of bond shown in the diagram below. [1]



3. A student suspects an unlabelled organic liquid is a carboxylic acid. Name the reagent(s) that must be added to the unknown organic liquid to test for the presence of a carboxylic acid. Give the expected observations for a positive result. [2]

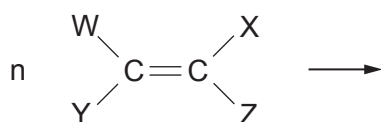
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4. Complete the equation below to show the product of addition polymerisation. [1]



5. State the meaning of the term 'standard enthalpy change of formation'. [2]

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